PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

GSD CarPass Depot App

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

"As university populations expand, the challenge of maintaining secure and organized parking facilities has transitioned from a basic logistical concern to a critical security imperative for modern school administrations. Efficient vehicle access control represents the foundational line of defense in campus safety, serving as a filter that ensures only verified personnel, students, and authorized visitors gain entry to the institutional perimeter. However, the current procedural framework for obtaining vehicle credentials often hinders this efficiency. Presently, the **carpass application workflow** requires a multi-stage physical transit: applicants must first report to the **General Services Department (GSD)** to initiate the process, proceed to the **Accounting Office** for payment verification, and finally return to the **GSD** for the issuance of the physical ticket. Under the prevailing system, an applicant must engage in a time-consuming physical 'circuit': first reporting to the GSD to initiate documentation, then migrating to the Accounting Office for manual payment processing, and finally returning to the GSD to present proof of payment and claim their physical ticket. This tethered movement between departments creates a dangerous administrative lag; it results in a period of 'unverified presence' where a vehicle may be operating on campus grounds long before its official registration is finalized in the system.

This administrative bottleneck is most acutely felt at the campus entry points. Because the issuance process is slow and manual, the subsequent **physical verification of tickets** at the gate becomes a labor-intensive hurdle. During peak rush hours, the time required for security personnel to manually check individual carpass tickets leads to **severe traffic congestion**, creating a 'choke point' at the university entrance. This not only frustrates the academic community but also creates a secondary security risk: as traffic backlogs grow, the pressure to maintain flow can lead to hurried inspections, further compromising the integrity of the perimeter.

At the University of Iloilo, the General Services Department (GSD) is tasked with the critical responsibility of managing vehicle entries and parking allocations. Currently, the university relies

on a traditional, manual car pass system where applicants are required to physically visit multiple offices, often multiple times. While this system has served its purpose in the past, the increasing volume of vehicles and the demand for faster transactions have made it obsolete. The manual verification of stickers at the gate often leads to morning traffic congestion, while the paper-based application process consumes valuable time for both the GSD staff and the faculty.

Furthermore, the lack of a centralized digital repository for vehicle data makes it difficult for the university to monitor real-time campus occupancy or quickly verify the authenticity of a car pass. In an era where digital transformation is optimizing every facet of society, the reliance on physical stickers and manual logbooks is becoming an outdated practice that hinders administrative productivity.

1.2 Statement of the Problem

Statement of the Problem

The current vehicle gate pass system at the University of Iloilo relies on manual processing, which creates several operational inefficiencies for both the General Services Department (GSD) and the faculty members. Specifically, the following problems have been identified:

- **Manual Application:** Applicants must physically visit the GSD office to submit documents and collect stickers, leading to lost time and administrative congestion.
- **Verification Delays:** Security personnel at the gates must manually verify physical car passes, which causes traffic build-up during peak hours.
- **Data Management Issues:** Maintaining paper-based records makes it difficult for the GSD to track the total number of registered vehicles, expire old passes, or quickly identify unauthorized vehicles on campus.
- **Security Risks:** Physical car passes are susceptible to being lost, damaged, or even forged, compromising the university's ability to maintain a secure parking environment.

1.3 Objectives of the Study

1.3.1 General Objective

To address these growing concerns, the GSD CarPass Depot App is proposed as a comprehensive digital solution. This mobile and web-based application aims to digitize the entire lifecycle of a car pass—from the initial application and document submission to real-time gate verification via QR code technology. By transitioning to a "depot" or a centralized digital system, the university can eliminate administrative bottlenecks, enhance campus security, and provide teachers with a modern, hassle-free experience. This project seeks to bridge the gap between traditional campus management and modern technological efficiency.

1.3.2 Specific Objectives

Specifically, the study aims to:

1. Implement QR Code Technology
2. Design a Security Scanning Feature
3. Establish a Centralized "Depot" Database
4. Enable Automated Notifications

1.4 Scope and Delimitation

Scope of the Study

The GSD CarPass Depot App focuses on:

- User Registration & Profile Management: A mobile and web interface for faculty, staff, and students to create accounts and register their vehicle details.
- Digital Document Submission: A module for users to upload clear photos or PDFs of their Official Receipt (OR) and Certificate of Registration (CR) for GSD review.
- GSD Admin Dashboard: A centralized "Depot" for the General Services Department to approve or reject applications, view real-time statistics, and manage the database of active vehicles.

- QR Code Generation: The system will generate a unique, non-transferable QR code for every approved vehicle, which serves as the digital "CarPass."
- Security Scanner Module: A dedicated mobile view for security guards to scan QR codes at the gate, instantly displaying the owner's photo, vehicle plate number, and pass status (Active/Expired).
- Automated Status Alerts: In-app notifications to remind users three days before their car pass expires.

Delimitation of the Study

To ensure the project remains manageable, the following are not included in the system:

- No Real-Time GPS Tracking: The app will not track the live location of vehicles once they are inside the campus; it only logs entry and exit times.
- No Payment Gateway: The app does not handle monetary transactions. Any fees for car passes must still be settled through the University Cashier (though users can upload a photo of the payment receipt).
- No Hardware Integration: The system will not physically open or close motorized gate barriers; it is a software-based verification tool for security personnel.
- No Traffic Violation Management: The app is for identification and entry purposes only; it does not record or process traffic tickets or parking fines.
- Limited to Campus Gates of University of Iloilo: The scanning feature is intended only for the university's main entry and exit points.

1.5 Significance of the Study

The proposed system will benefit the following:

- **The General Services Department (GSD):** This study will provide the GSD with a centralized "depot" or digital database, eliminating the need for manual

record-keeping and physical filing. It allows for faster processing of applications and real-time monitoring of all registered vehicles on campus.

- **Faculty and Staff:** Teachers will benefit from a more convenient application process. Instead of physically visiting the GSD office and waiting in line, they can submit their documents digitally, saving time and allowing them to focus on their primary academic duties.
- **Students:** Student drivers will experience a faster and more professional registration process. A digital pass ensures they can enter the campus efficiently, reducing the risk of being late for classes due to gate congestion.
- **University Security Personnel:** This project provides security guards with a reliable mobile tool to verify vehicle authenticity instantly. It removes the guesswork involved in checking physical stickers, especially during nighttime or bad weather, thereby improving overall campus safety.

1.6 Definition of Terms

- **GSD:** General Services Department. an administrative division responsible for maintaining a safe, clean, and functional learning environment
- **Department Secretary:** Refers to the personnel from the General Services Department authorized to review, approve, or reject vehicle registration applications through the system dashboard.
- **CarPass Depot:** The centralized digital database and storage system where all vehicle records, owner information, and document logs are securely kept and managed.
- **Digital CarPass:** A virtual permit generated by the app in the form of a unique QR code, serving as the official electronic substitute for traditional physical windshield stickers.
- **Gate Verification:** The process of scanning a user's QR code at the university entrance to confirm their identity and the validity of their vehicle's access rights.
- **User Behavior Monitoring** – Tracking and analyzing user actions to ensure safety and compliance.
- **Security Scanner Module:** A specific feature of the app designed for security personnel to read QR codes and instantly view the "Active" or "Expired" status of a vehicle.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

The use of digital technologies in access control has improved monitoring efficiency and security in institutional environments. Jaiswal, Singh, and Thusu (2023) presented a smart and secure gate pass system using QR codes to automate entry and exit monitoring. Their study explained that QR-based verification reduces paperwork, improves record management, and speeds up identity checking through digital scanning linked to a database.

In a related study, Hamid, Gee, Bahaman, Anawar, Ayob, and Malek (2018) developed an intelligent automated gate system that uses QR-code passes for authorized staff. Their system was designed to record incoming and outgoing activities, trigger alerts for unauthorized access, and improve gate security through internet-connected monitoring. The study emphasized that QR-based gate automation can provide a practical and lower-cost alternative for organizations that need secure but affordable access control.

Yu, Islam, and Thong (2022) also proposed an enhanced QR-code-based smart parking system for a mobile environment. Their study showed that users could reserve parking spaces, scan QR codes to begin parking sessions, and manage transactions through a smartphone application. The authors highlighted that QR-supported parking systems can reduce congestion, save time, and improve the efficiency of parking operations in controlled environments such as universities.

Recent research by Johanna (2025) highlights how integrating QR codes with Optical Character Recognition (OCR) can significantly improve vehicle access systems. Their study found that

using QR codes for both entry and exit validation replaces the need for manual cards and reduces operational costs while maintaining over 90% accuracy in vehicle identification.

Current global trends show a shift toward "dematerialized" or fully digital vehicle registration certificates. Starting in 2025, many administrative systems are phasing out paper formats in favor of digital versions to reduce fraud, as digital documents are considered "forgery-proof" compared to physical stickers.

2.2 Local Studies

In the Philippine setting, Fernandez (2022) developed an RFID-based monitoring vehicle management information system with SMS notification. The study described a system where registered vehicles use RFID tags for gate verification, while the database records vehicle movement and sends notifications related to tag validity. Results showed excellent ratings in functionality, usability, and performance, indicating that automated vehicle monitoring can strengthen security while reducing manual checking at gates.

Similarly, Mimbala, Macre, Maunga, and Acala (2024) developed a web-based vehicle monitoring system for Mindanao State University–Lanao del Norte Agricultural College. Their study replaced manual logbook recording with QR-code verification and SMS notification, resulting in improved accuracy, reduced processing time, and better security in monitoring campus vehicle entry and exit. The system also allowed administrators to generate reports and maintain digital vehicle records more efficiently.

Another Philippine-related study by Panganiban and Dela Cruz (2017) described an RFID-based vehicle monitoring system that used RFID, GSM, sensors, and a barrier gate to manage vehicle entry. The study reported that the system satisfied functional requirements and was perceived as usable and reliable, showing that automated vehicle monitoring technologies are feasible for local institutional applications.

These local studies show that Philippine institutions benefit from replacing manual vehicle logging with digital verification systems because such systems improve record accuracy, speed up processing, and provide better control over authorized access.

2.3 Related Literature

QR code technology is widely used in identification and access systems because it can store information efficiently and can be read quickly using scanners or smartphones. In secure QR applications, verification mechanisms can also be combined with cryptographic methods to protect data integrity and authenticity. Bani-Hani, Wahsheh, and Al-Sarhan (2014) described a secure QR code system that used digital-signature-based verification to help ensure the authenticity of QR code contents. This shows that QR codes are not only useful for speed and convenience but can also support stronger security when properly implemented.

Web-based information systems are also important in modern organizational operations because they support data processing, monitoring, reporting, and decision-making. Laudon and Laudon's Management Information Systems: Managing the Digital Firm explains that information systems help organizations transform data into meaningful information that improves business processes and management performance. In the context of vehicle access monitoring, this supports the use of centralized databases and web dashboards for storing records, verifying users, and generating reports.

Digital vehicle monitoring systems also help reduce the limitations of manual procedures. Manual logbooks are often slow, error-prone, and difficult to retrieve for reporting. By contrast, automated systems can validate users in real time, maintain updated records, and improve communication between users and administrators through notifications and digital logs. These capabilities are consistently emphasized in studies on smart parking and automated vehicle monitoring.

In addition, mobile-based scanning technology makes QR systems practical in field settings. Guards or security personnel can use a scanner or smartphone to verify access instantly, making the entry and exit process faster and more convenient while preserving security. This is especially useful in campuses, offices, and depot environments where vehicle movement must be monitored continuously.

2.4 Synthesis of the Reviewed Literature

The reviewed studies and literature show that digital technologies such as QR codes, RFID, SMS notification, mobile applications, and web-based information systems significantly improve vehicle monitoring, access control, and operational efficiency. Foreign studies demonstrate that QR-based systems can automate gate management, improve parking operations, and strengthen security by verifying authorized users quickly and accurately.

Local studies further confirm that these technologies are effective in the Philippine context. Research conducted in local institutions showed that RFID- and QR-based vehicle monitoring systems reduce manual workload, increase accuracy, improve report generation, and provide better control over vehicle entry and exit. These findings indicate that automated access systems are practical and beneficial for institutional use in the country.

The literature also highlights the importance of centralized databases, real-time verification, and user-friendly system design. These features ensure that authorized users can be validated efficiently while administrators retain accurate digital records for monitoring and reporting. Secure QR implementation further strengthens system reliability by helping prevent tampering or unauthorized access.

Overall, the reviewed literature strongly supports the development of the GSD CarPass Depot App. The proposed system is aligned with existing studies that emphasize automation, security, accuracy, and efficient vehicle access management. By integrating QR code verification with a web-based monitoring platform, the proposed system can provide a reliable and modern solution for controlling and recording vehicle movement in a depot environment.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study utilizes a developmental research design, which focuses on the systematic design, development, and evaluation of the GSD CarPass Depot App. Developmental research is appropriate for this study because it aims to create a functional system that addresses real-world problems encountered by the General Services Department (GSD) of the University of Iloilo.

The study involves analyzing the existing manual car pass process, identifying its limitations, and developing a digital solution that improves efficiency, security, and data management. The system will then be evaluated based on its functionality, usability, and performance.

3.2 System Development Methodology

The study adopts the Agile Development Model, a flexible and iterative approach that allows continuous improvement of the system through repeated cycles.

The development process includes the following phases:

1. Requirements Analysis

In this phase, the researchers gathered information about the current manual car pass system through interviews and observations. The needs of GSD staff, security personnel, and users (faculty, staff, and students) were identified to define system requirements.

2. System Design

The system architecture, database structure, and user interface were designed. This includes the design of:

- User registration module
- QR code generation system
- Admin dashboard
- Security scanning module

Diagrams such as Data Flow Diagrams (DFD), Entity Relationship Diagrams (ERD), and Use Case Diagrams were also created.

3. Development

The actual coding and development of the system were performed using selected programming languages and tools. The modules were developed incrementally, ensuring that each feature meets the required specifications.

4. Testing

Each module of the system was tested to identify errors and ensure proper functionality. Testing includes unit testing, integration testing, and system testing.

5. Deployment (Prototype Implementation)

The developed system was deployed as a prototype and tested in a simulated or actual environment within the university to evaluate its effectiveness.

3.3 System Architecture

The GSD CarPass Depot App follows a client-server architecture.

- **Client Side:**

  Users (students, faculty, staff, and security personnel) interact with the system through a mobile or web interface.

- **Server Side:**

  The server processes requests, validates data, generates QR codes, and manages database transactions.

- **Database:**

  A centralized database stores user profiles, vehicle details, QR codes, and access logs.

This architecture ensures real-time data access, efficient processing, and secure management of information.

3.4 Tools and Technologies Used

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python
- **Database:** MySQL
- **Development Tools:** Visual Studio Code, Workbench

3.5 Data Gathering Techniques

The researchers used the following methods to collect data:

- **Interviews:**
  Conducted with GSD personnel and security guards to understand the current system and its challenges.

- **Observation:**
  Direct observation of the manual car pass process, including application, verification, and gate entry procedures.

- **Document Analysis:**
  Review of existing forms, logs, and procedures used in the current system.

These techniques ensured that the system requirements were accurate and aligned with actual operational needs.

3.6 Testing Procedures

To ensure system quality, the following testing procedures were conducted:

- **Unit Testing:**
  Individual components (e.g., QR generation, login system) were tested independently.

- **Integration Testing:**
  Combined modules were tested to ensure proper interaction between components.

- **System Testing:**
  The entire system was tested to verify that it meets all functional requirements.

- **User Acceptance Testing (UAT):**
  Selected users (GSD staff and security personnel) tested the system to evaluate usability and effectiveness.

3.7 Evaluation Criteria

The system was evaluated based on the following criteria:

- **Functionality:**
  The system's ability to perform required tasks such as registration, QR generation, and verification.

- **Usability:**
  Ease of use for both administrators and users.

- **Efficiency:**
  Speed of processing applications and verifying vehicle access.

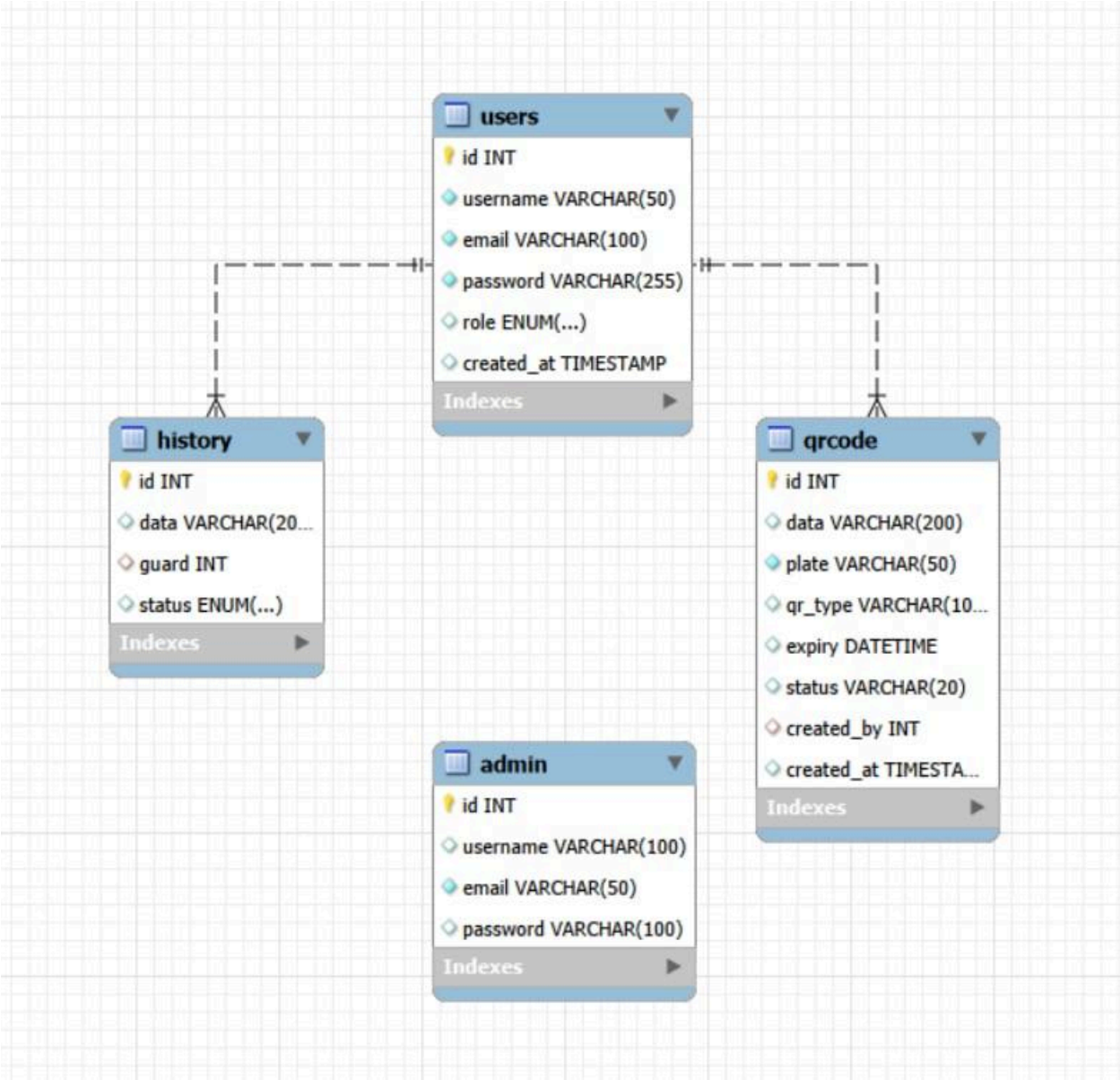
- **Security:**
  Ability to prevent unauthorized access and protect user data.

- **Reliability:**
  Consistency of system performance during operation.

SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD—Level 0)

Entity Relationship Diagram (ERD) Entities:



Relationships:

- User Register the form after pay in the teller send the prof of payment and Generate the carpass QRcode renew the car pass when its Expire
- Guard check the carpass QRcode IN and OUT generate the QR for visitors
- GSD Admin Approve and Monitor the User and Visitors

Use Case Diagram

Actors:

- User
- Grauds
- GSD Admin

Use Cases:

- Register Account
- Pay to get a Carpass
- Renew a Carpass
- Monitor the In and Out of the user
- Generate the QR carpass for the Visitors
- Manage Accounts (Admin)
- Approve the Carpass and Renewal (Admin)

End of Proposal

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.